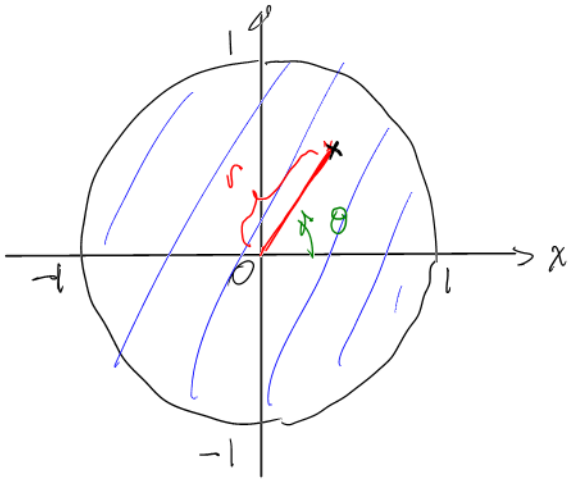
19. Compute $\frac{1}{\pi}\iint_R (3+x^4+2x^2y^2+y^4)\,dA$ where R is the closed disk of unit radius centered at the origin. Give the answer as an integer or a fraction in lowest terms.

$\frac{10}{3}$



Polar coords: $x = r\cos\theta$, $y = r\sin\theta$

$dA = dx\,dy = r\,dr\,d\theta$

$$\frac{1}{\pi}\iint_R (3+x^4+2x^2y^2+y^4)\,dA$$

$$= \frac{1}{\pi}\int_{\theta=0}^{\theta=2\pi}\int_{r=0}^{r=1}(3+r^4)\,r\,dr\,d\theta$$

$$= \left(\frac{1}{\pi}\int_{\theta=0}^{\theta=2\pi} d\theta\right)\left(\int_{r=0}^{r=1}(3+r^4)\,r\,dr\right)$$

$\left(\frac{1}{\pi}\int_{\theta=0}^{\theta=2\pi} d\theta\right) = 2$

$\left[\frac{3r^2}{2} + \frac{r^6}{6}\right]_{r=0}^{r=1} = \frac{3}{2} + \frac{1}{6} = \frac{5}{3}$

$(3+x^4+2x^2y^2+y^4)$

$= (x^2+y^2)^2 = r^4$

$3 + (r\cos\theta)^4 + 2(r\cos\theta)^2(r\sin\theta)^2 + (r\sin\theta)^4$

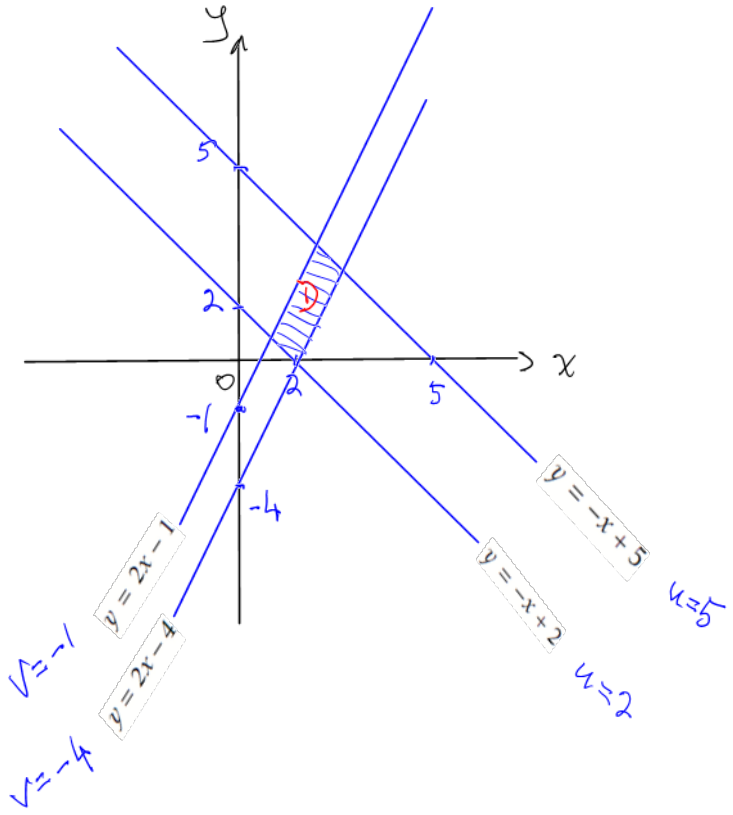
$= \cdots$ simplify.

$= \frac{10}{3}$

20. Let D be the parallelogram in the xy-plane bounded by the lines $y = -x + 5$, $y = -x + 2$, $y = 2x - 1$, $y = 2x - 4$. Compute the integral $\iint_D e^x \, dydx$.

Give the answer as an expression in terms of integer powers of e.

$$\boxed{3e(e-1)^2} = 3e^3 - 6e^2 + 3e^1$$



Change of variables:

$$(u,v) \in \mathbb{R}^2 \xrightarrow{\Phi} (x,y) \in \mathbb{R}^2 \xrightarrow{f(x,y) = e^x} \mathbb{R}$$

$\Phi^{-1}(D)$ $\quad u = x+y$, $\quad v = y-2x$ $\quad D$

$u - v = 3x$

$$J_\Phi(u,v) = \frac{\partial(x,y)}{\partial(u,v)} = \det\begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix}$$

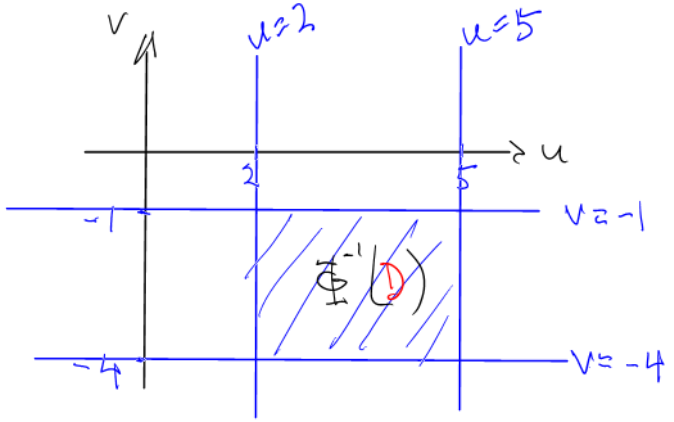
$$J_{\Phi^{-1}}(x,y) = \frac{\partial(u,v)}{\partial(x,y)} = \det\begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}$$

$$= \det\begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix} = 3$$

Know: $J_\Phi(u,v) = \dfrac{1}{J_{\Phi^{-1}}(x,y)} = \dfrac{1}{3}$

const indep of x, y

Φ $\quad \Phi^{-1}$ $\quad u = x+y$, $\quad v = y-2x$



$$\iint_D e^x \, dy\,dx = \iint_{\Phi^{-1}(D)} e^{\frac{1}{3}(u-v)} \frac{\partial(x,y)}{\partial(u,v)} \, du\,dv$$

$$= \int_{v=-4}^{v=-1} \int_{u=2}^{u=5} e^{u/3} \cdot e^{-v/3} \, \frac{1}{3} \, du\,dv$$

$$= \left(\int_{v=-4}^{v=-1} e^{-v/3} \, dv \right) \left(\int_{u=2}^{u=5} e^{u/3} \cdot \frac{1}{3} \, du \right)$$

$$\left[-3e^{-v/3} \right]_{v=-4}^{v=-1} \qquad \frac{1}{3} \left[3e^{u/3} \right]_{u=2}^{u=5}$$

$$= -3\left(e^{1/3} - e^{4/3}\right) \qquad = \left(e^{5/3} - e^{2/3}\right)$$

$$= 3\left(e^{9/3} - e^{6/3} - e^{6/3} + e^{3/3}\right)$$

$$= 3\left(e^3 - 2e^2 + e^1\right) = 3e\left(e-1\right)^2$$

21. Let R be the region in first quadrant of the xy-plane bounded by the curves $x^2+y^2=9$, $x^2+y^2=16$ and the lines $y=2x+1$, $y=2x+2$. Compute the integral

$$\iint_R (x+2y)(x^2+y^2)\,dydx.$$

Give the answer as an integer or a fraction in lowest terms.

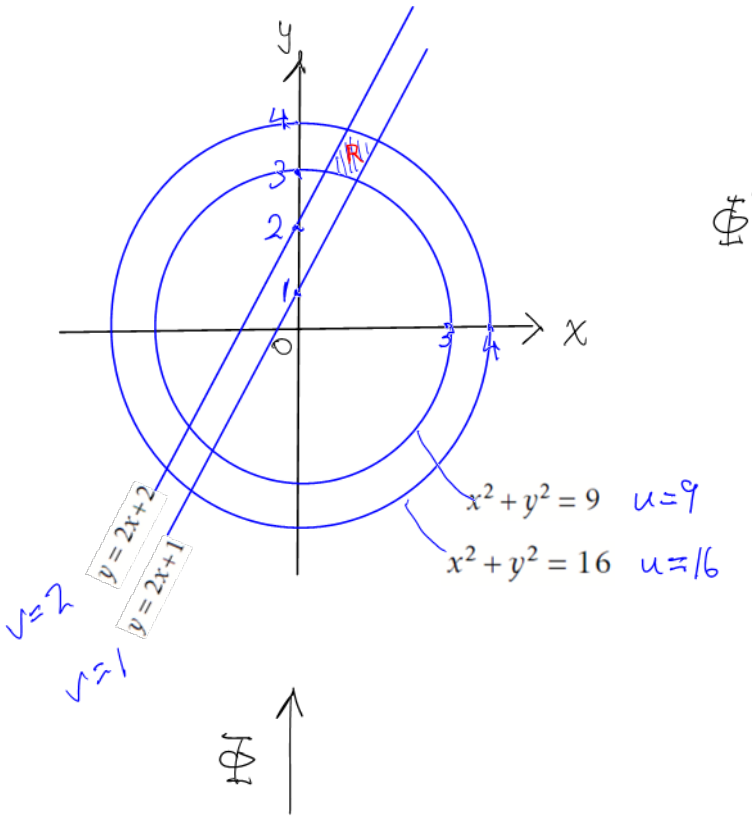
$\frac{175}{4}$

Change of variables

$$(u,v) \xrightarrow{\Phi} (x,y) \xrightarrow{f(x,y)=(x+2y)(x^2+y^2)} \mathbb{R}$$

$(u,v) \in \mathbb{R}^2$ $\quad (x,y) \in \mathbb{R}^2$

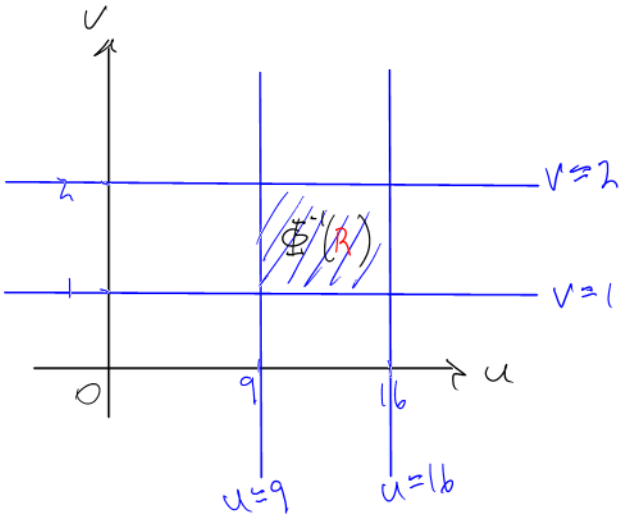
$\Phi^{-1}(R)$ $\quad u = x^2+y^2$, $v = y-2x$



$$J_\Phi(u,v) = \frac{\partial(x,y)}{\partial(u,v)} = \det\begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix}$$ hard to compute

$$J_{\Phi^{-1}}(x,y) = \frac{\partial(u,v)}{\partial(x,y)} = \det\begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix}$$ easy to compute

$$= \det\begin{pmatrix} 2x & 2y \\ -2 & 1 \end{pmatrix} = 2x+4y$$



So $$\iint_R (x+2y)(x^2+y^2)\,dy\,dx$$

$$= \iint_{\Phi^{-1}(R)} (x+2y)(\underbrace{x^2+y^2}_{u})\,\underbrace{\frac{\partial(x,y)}{\partial(u,v)}}_{\frac{1}{\left|\frac{\partial(u,v)}{\partial(x,y)}\right|}}\,du\,dv = \frac{1}{2(x+2y)}\,du\,dv$$

$$= \int_{v=1}^{v=2}\left(\int_{u=9}^{u=16} u\cdot\frac{1}{2}\,du\right)dv$$

$$= \int_{v=1}^{v=2}\left[\frac{u^2}{4}\right]_{u=9}^{u=16} dv = \int_{v=1}^{v=2}\frac{1}{4}(256-81)\,dv$$

$$= \frac{1}{4}(256-81) = \frac{175}{4}$$

22. Consider the double integral

$$\iint_R \frac{y^2}{x} e^{x^2}\, dA$$

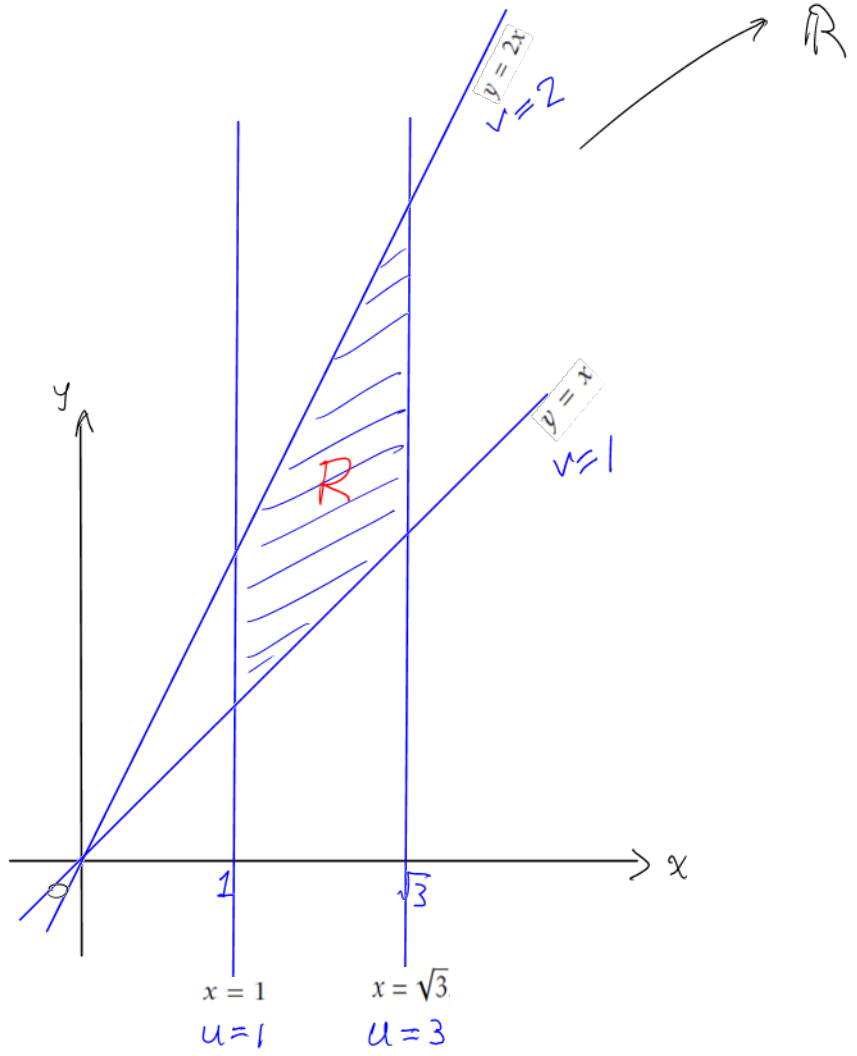
where R is the region in the first quadrant of the xy-plane bounded by the curves $x = 1$, $x = \sqrt{3}$, and the lines $y = x$, $y = 2x$. Use the change of variables $u = x^2$ and $v = \dfrac{y}{x}$ to rewrite the above double integral as an iterated integral in the variables u, v and with integration order $du\, dv$.

[You do not have to compute the integral.]

Give the answer in the form $\displaystyle\int_a^b \int_c^d g(u,v)\, du\, dv$, where a and b are real numbers, c and d are real numbers or expressions in terms of v, and $g(u,v)$ is an expression in terms of u and v.

$$\int_{v=1}^{v=2} \int_{u=1}^{u=3} \frac{1}{2} u^{1/2} v^2 e^u \, du\, dv$$

$f(x,y) = \dfrac{y^2}{x} e^{x^2}$



$(x,y) \in \mathbb{R}^2$ R

$$J_{\Phi}(u,v) = \frac{\partial(x,y)}{\partial(u,v)} = \det\begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix}$$

$$= \det\begin{pmatrix} \frac{1}{2}u^{-1/2} & 0 \\ \frac{1}{2}u^{-1/2}v & u^{1/2} \end{pmatrix}$$

$$= \frac{1}{2} \quad \text{const!}$$

Φ

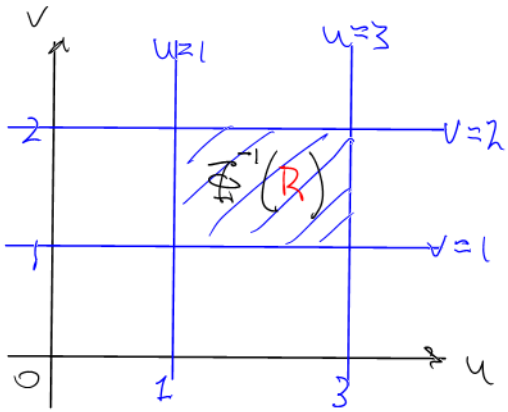
$(u,v) \in \mathbb{R}^2$

Ch. of Var.

$u = x^2, \quad v = \frac{y}{x}$

$\Updownarrow$

$x = \sqrt{u}, \quad y = \sqrt{u}\cdot v$



$$\iint_R \frac{y^2}{x} e^{x^2}\, dA = \iint_{\Phi^{-1}(R)} \underbrace{u^{1/2}v^2 \cdot e^u}_{\frac{y^2}{x}e^{x^2}} \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv$$

$dA = dx\,dy$

$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv = \frac{1}{2}\, du\,dv$

$$= \int_{v=1}^{v=2} \int_{u=1}^{u=3} \frac{1}{2} u^{1/2} v^2 e^u \, du\,dv$$

$$\int_a^b \int_c^d g(u,v)\, du\, dv$$

23. Compute $\int_{-2}^{2}\int_{-2}^{2}\int_{-2}^{2} xz^2\,dzdydx$.

Give the answer as an integer or a fraction in lowest terms.

$$\int_{-2}^{2}\int_{-2}^{2}\int_{-2}^{2} xz^2\,dzdydx$$

$$= \int_{x=-2}^{x=2}\left(\int_{y=-2}^{y=2}\left(\int_{z=-2}^{z=2} xz^2\,dz\right)dy\right)dx$$

$$= \int_{x=-2}^{x=2}\left(\int_{y=-2}^{y=2} x\left[\frac{z^3}{3}\right]_{z=-2}^{z=2} dy\right)dx$$

$$= \int_{x=-2}^{x=2}\left(\int_{y=-2}^{y=2} x\cdot\frac{16}{3}\,dy\right)dx$$

$$= \int_{x=-2}^{x=2}\frac{16}{3}x\cdot 4\,dx = \left[\frac{32}{3}x^2\right]_{x=-2}^{x=2}$$

$$= \frac{32}{3}\cdot\left(2^2-(-2)^2\right) = 0$$

0

Easier: simply observe that the region of $\int$-tion is symmetric wrt reflecting about 0

$(x,y,z)\mapsto(-x,-y,-z)$

but the integrand xz^2 is antisymmetric

$\Rightarrow \int$ is 0

24. Compute the integral

$$\int_0^1 \int_0^x \int_{x^2+y^2}^2 dz\,dy\,dx.$$

Give the answer as an integer or a fraction in lowest terms.

$\frac{2}{3}$

$$\int_0^1 \int_0^x \int_{x^2+y^2}^2 dz\,dy\,dx = \int_{x=0}^{x=1} \int_{y=0}^{y=x} \int_{z=x^2+y^2}^{z=2} dz\,dy\,dx$$

$$= \int_{x=0}^{x=1} \int_{y=0}^{y=x} (2-x^2-y^2)\,dy\,dx$$

$$= \int_{x=0}^{x=1} \underbrace{(2-x^2)\cdot x}_{2x-x^3} - \underbrace{\left[\frac{y^3}{3}\right]_{y=0}^{y=x}}_{\frac{x^3}{3}}\,dx = \int_{x=0}^{x=1} 2x - \frac{4}{3}x^3\,dx$$

$$= \left[x^2 - \frac{1}{3}x^4\right]_{x=0}^{x=1} = 1 - \frac{1}{3} = \frac{2}{3}$$